Distributed Edge Artificial Intelligence

A Thesis (Major Project) submitted in partial fulfilment of the requirements

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

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2026

Declaration

We, the students of Computer Science & Engineering, Barak Valley Engineering College, declare that the project report entitled **"Distributed Edge Artificial Intelligence"** has been prepared under the guidance of **Mr. N Rana Singha**, Computer Science & Engineering, Barak Valley Engineering College.

The report has been prepared without resorting to plagiarism. We have adhered to the principles of academic honesty and integrity. No falsified or fabricated data have been presented in this report. Further, the matter embodied in this project report has not been submitted previously for the award of any degree or diploma.

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Abstract

Distributed Edge Artificial Intelligence (Edge AI) places artificial intelligence workloads across devices near data sources, edge servers, and cloud resources. This report presents a preliminary literature study of how such workloads can be distributed while accounting for communication, computing capacity, energy use, and privacy. A thematic review is used to examine end-edge-cloud architectures, federated learning, collaborative inference, communication efficiency, resource heterogeneity, and security considerations. The reviewed work indicates that distributing computation closer to data can support responsive applications and reduce dependence on centralised processing, but the benefits depend on task placement, network conditions, and device capabilities. Across the four selected works, communication is a recurring constraint, but federated training, task scheduling, and partitioned inference address different problems and use different evaluation settings. The project remains at the literature-review and feasibility-assessment stage; no implementation or experimental measurements are claimed. Future work will select an application and platform, then evaluate an appropriate distributed-AI design using reproducible performance and energy measures.

Keywords: Distributed Edge AI, Edge Intelligence, Federated Learning, Collaborative Inference, End-Edge-Cloud Computing

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Chapter 1

Introduction

Artificial intelligence (AI) is increasingly used in applications that process data generated by mobile devices, sensors, and Internet of Things (IoT) systems. Sending all such data to a distant cloud for analysis can require sustained network communication and may not suit applications with limited bandwidth or strict response-time requirements. These pressures have encouraged interest in performing at least part of the computation closer to where the data is produced.

Distributed Edge Artificial Intelligence studies how AI workloads can be placed across resource-constrained end devices, nearby edge servers, and cloud infrastructure. Rather than treating every device as an independent system, distributed approaches allow multiple nodes to contribute to training, inference, or other application tasks. The resulting design must decide which tasks to execute at each layer, what information to exchange, and how to coordinate nodes with different capabilities.

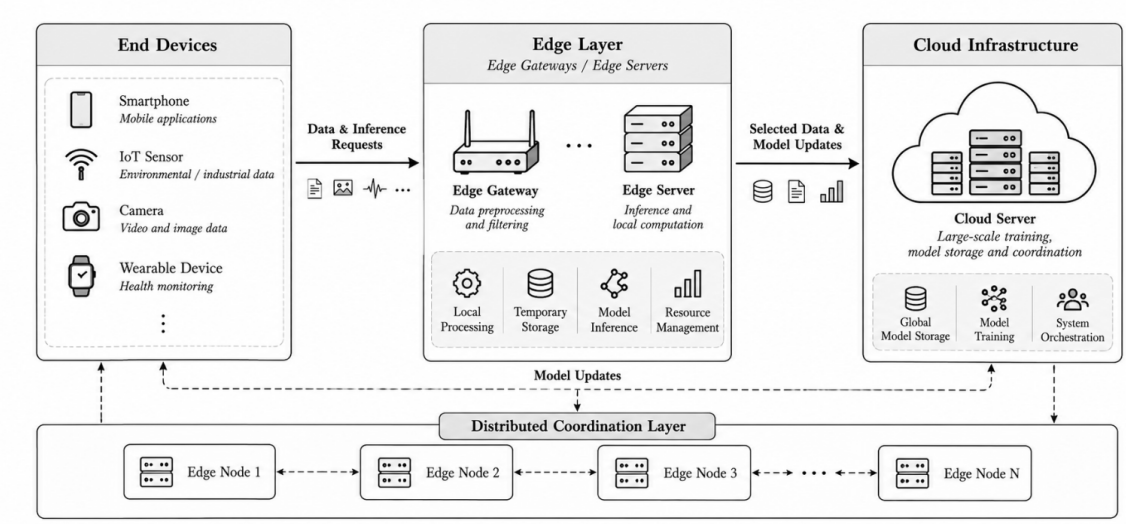


Figure 1.1: Conceptual Overview of Distributed Edge Artificial Intelligence.

1.1 Background

Edge AI refers to executing artificial intelligence computations close to the data source, at the network edge, rather than in a centralised cloud data centre [2]. Edge resources may include end-user devices, gateways, and nearby edge servers, working alongside centralised cloud resources. These layers differ in proximity to users, compute capacity, storage, and network connectivity. A system may therefore allocate lightweight or latency-sensitive work to nearby devices and use more powerful resources for tasks that exceed their local

capacity.

Figure 1.2 illustrates how end devices, edge resources, cloud infrastructure, and applications can participate in a distributed Edge AI system.

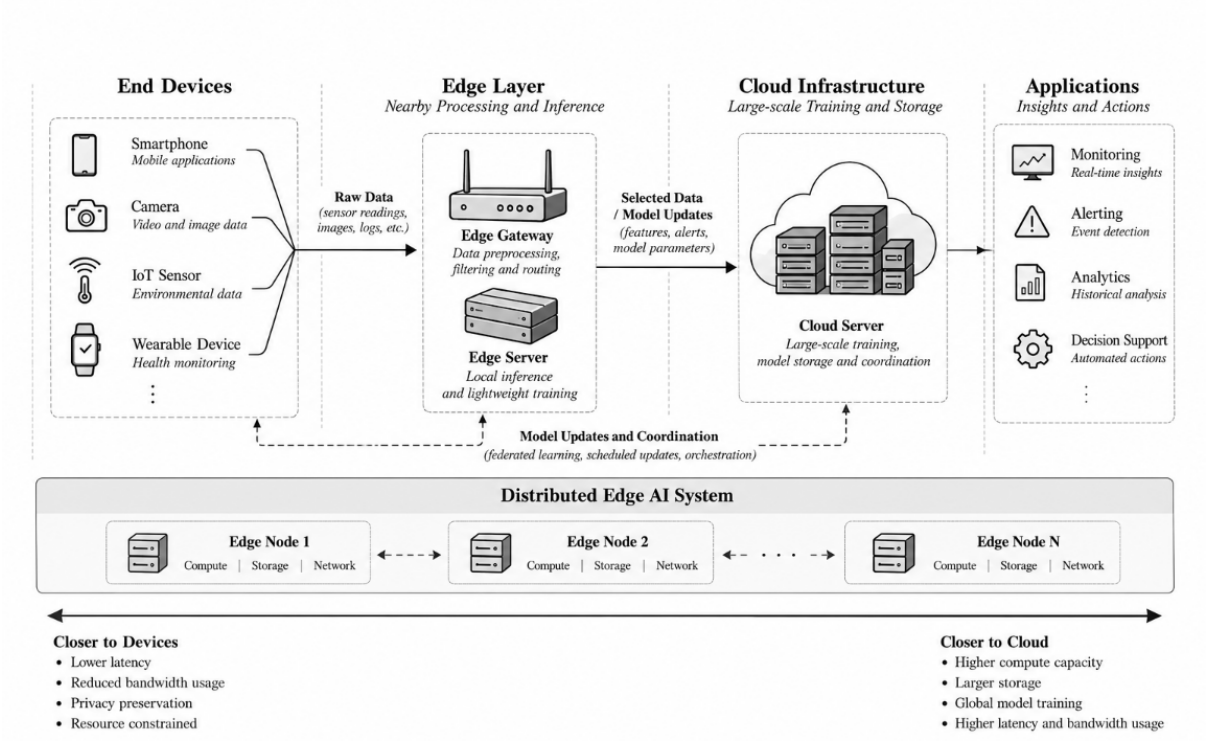


Figure 1.2: Background of Edge AI and Its Role in a Distributed Architecture.

Edge intelligence encompasses both the use of AI to manage edge systems and the execution of AI workloads on edge resources [2]. Distributed Edge AI extends this setting by coordinating work across interconnected, heterogeneous nodes. For example, a distributed scheduler can place application tasks across edge nodes as system conditions change [1]. In collaborative inference, a model's computation can be partitioned so that multiple devices contribute to producing a result [3]. These approaches make task placement and coordination important parts of the system design.

The choice of where to execute a workload affects more than processing speed. Sending data or intermediate model outputs across a network consumes bandwidth and introduces communication delay; keeping computation local can reduce some transfers but is constrained by device capacity and energy availability. Federated learning offers a way for clients to train collaboratively without collecting their raw training data at one central location, while still requiring the exchange and coordination of model updates [2]. Partitioned inference, in turn, must balance local computation against the cost of transferring intermediate results [3].

There is no single placement strategy that fits every workload. A design that is appropriate for a delay-sensitive application may not be optimal when energy, network availability, or device diversity is the main constraint. Distributed systems must account

for communication cost, heterogeneous computing resources, changing node availability, and the privacy and security implications of sharing data or model information [2, 1]. This report reviews these considerations as a basis for selecting a suitable application and implementation direction in a later project phase.

1.2 Problem Statement

The central problem considered in this project is how to distribute AI training and inference across end devices, edge resources, and cloud infrastructure when those resources have different capabilities and communication links have a cost. Centralised execution can require data to travel away from its source, whereas fully local execution may exceed the computing or energy capacity of an individual device. Distributing work therefore introduces trade-offs among response time, communication, resource use, and coordination.

This report examines those trade-offs through a preliminary literature review and feasibility assessment. It focuses on end-edge-cloud architectures, federated training, collaborative inference, communication efficiency, resource heterogeneity, and privacy and security considerations. The specific application and implementation platform have not yet been selected, and no system has been implemented or experimentally evaluated as part of this stage. The review is intended to identify relevant design choices and provide a grounded basis for defining the next phase of work. It is guided by two questions: how do federated training and distributed inference differ in their methods and system requirements, and which measurable constraints should determine whether a candidate approach is feasible on the resources available to the team?

1.3 Objectives

The objectives of this preliminary study are:

1. Review distributed edge-AI architectures and approaches to federated training and collaborative inference.

2. Identify key challenges related to communication, heterogeneity, privacy, energy, and available hardware resources.

3. Establish a feasibility-assessment framework to guide selection of an application and platform for future implementation and evaluation.

1.4 Scope and Report Organization

The scope of this report is a literature-stage study of distributed edge AI; it does not claim a completed implementation, selected application, or new experimental measurements. The review considers architecture, training, inference, communication, heterogeneity, privacy and security, and hardware and energy constraints. Chapter 2 surveys the related literature. Chapter 3 describes the review protocol and proposed feasibility-assessment and technical approaches. Chapter 4 presents the literature synthesis and discusses feasibility and limitations. Chapter 5 summarises the conclusions and identifies future work.

The project follows a six-phase research and development process: (1) learning the fundamentals of Edge AI and Distributed Edge AI, (2) reviewing research papers and building a verified reference collection, (3) conducting small-scale experiments, (4) assessing the feasibility of a larger project, (5) implementing the selected system, and (6) testing and refining the implementation. At the current stage, the work focuses on the first two phases. The subsequent phases will be undertaken as understanding of the field develops and a suitable application is identified. The overall plan and current stage are summarised in Figure 1.3.

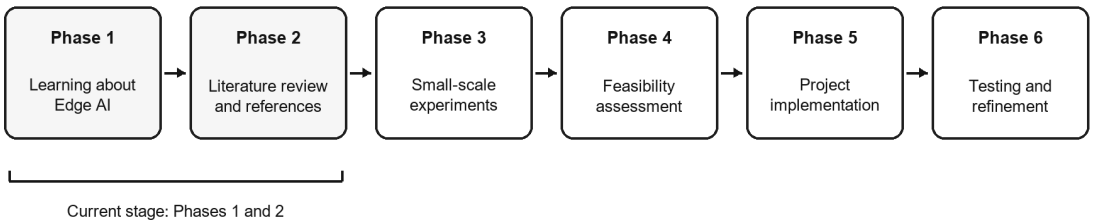


Figure 1.3: Six-Phase Research and Development Plan.

Chapter 2

Literature Survey

2.1 From Cloud AI to Edge Intelligence

The reviewed literature positions distributed edge AI as a response to the limits of centralised cloud processing. Edge AI performs AI computation near the user at the network edge rather than in a centralised data centre; this shift has been enabled by improvements in AI efficiency, the proliferation of IoT devices, and the growth of edge computing. Zhou et al. describe edge intelligence as extending AI to the "last mile" for mobile and IoT devices that generate data at the edge. Deng et al. distinguish AI for edge from AI on edge, with the latter covering model training and inference on edge devices [2]. The term "edge" has no universally accepted definition; its relation to cloud computing depends on proximity, capacity, and device heterogeneity [2].

2.2 End-Edge-Cloud Architectures

A common architectural pattern distributes processing across end devices, edge servers, and the cloud. End devices handle lightweight local computation, while edge servers provide nearby, more capable resources. The literature distinguishes the complementary roles of AI for edge and AI on edge, and describes orchestration of heterogeneous on-device, edge, and cloud resources for distributed AI. Collaborative learning across cloud and edge resources is also studied [2]. Hu et al. describe a distributed edge-AI scheduler that places application tasks across heterogeneous nodes while accounting for changing system conditions [1]. The layers and their data and model flows are summarised in Figure 1.1, which shows example end devices, edge resources, cloud services, and coordination among edge nodes. The arrows distinguish communication between the layers and the flow of data, inference requests, and model updates.

2.3 Distributed Training: Federated Learning

Federated learning (FL) trains local models on multiple clients without collecting raw user data at a central data centre. Federated learning and knowledge distillation are described as training approaches, while model splitting and partitioning support inference. FL challenges include statistical heterogeneity, communication cost, resource allocation, and privacy and security. Communication can dominate computation when local datasets are small and participant processors are comparatively fast [2]. Figure 2.1 illustrates the

local-training and model-aggregation workflow.

Communication efficiency, resource allocation, and data heterogeneity are recurring concerns in distributed edge learning [2]. McMahan et al. present federated averaging, in which clients perform local training and a server averages their model updates. Their evaluation covers five model architectures and four datasets, and reports 10–100 times fewer communication rounds than synchronized stochastic gradient descent in the settings tested [4]. This result is specific to their workloads and baselines; it does not imply the same reduction for every edge-learning system.

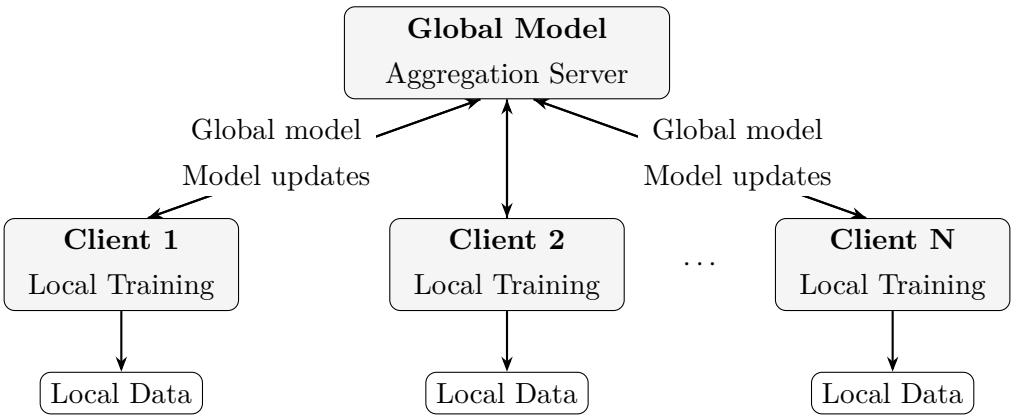


Figure 2.1: Federated Learning Workflow. Adapted from Deng et al. [2].

2.4 Distributed Inference: Partitioning and Collaboration

Model-partition methods split a deep neural network between a device and server based on transmission latency and device energy. Early-exit methods can reduce communication when intermediate features are sufficient. DistrEdge applies deep reinforcement learning to distribute CNN inference across heterogeneous IoT edge devices and reports a 1.1–3× speedup over the state-of-the-art baselines evaluated in that study; this is a study-specific result rather than a general speedup guarantee [3]. The device–server split is shown in Figure 2.2.

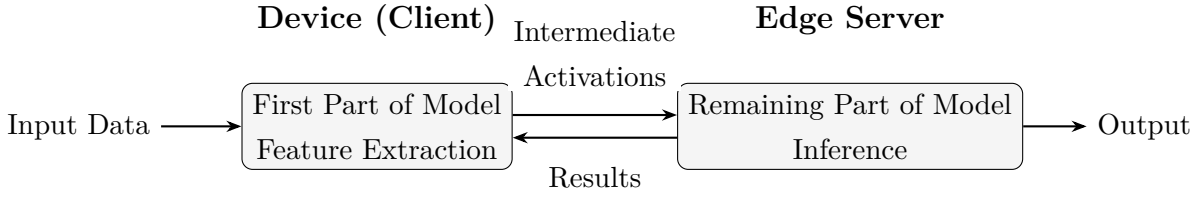


Figure 2.2: Partitioned CNN Inference between a Device and Edge Server. Adapted from Hou et al. [3].

Hu et al. also describe a video-object-detection application divided into frame extraction,

object detection, and frame combination services [1]. Figure 2.3 shows this pipeline as a second example of distributed inference. Assigning detection work to multiple containers can use available parallel capacity and reduce frame-processing time up to a point. However, frames and detection results must move between pipeline stages, adding network traffic and communication delay; the reported benefit depends on the video quality, container count, and deployment evaluated by Hu et al. [1].

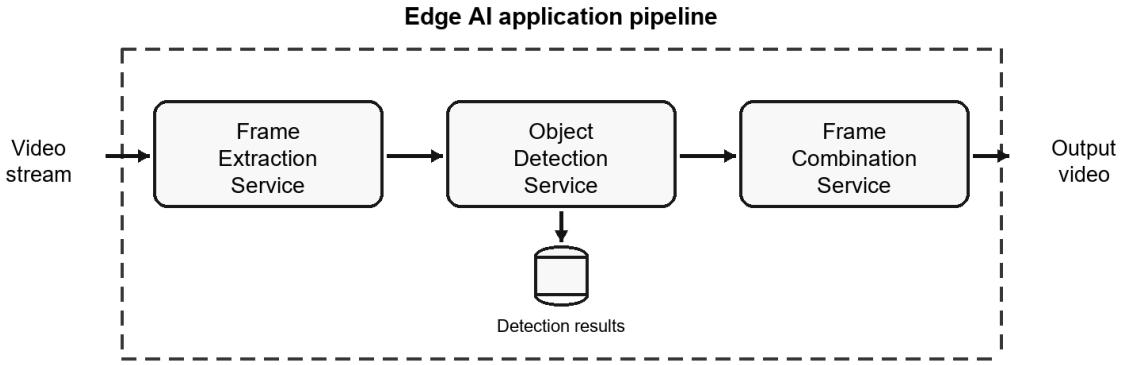


Figure 2.3: Distributed Video Object-Detection Pipeline. Adapted from Hu et al. [1].

2.5 Communication Efficiency

Communication is a central bottleneck across training and inference. What is transmitted affects per-round overhead and the number of communication rounds. Quantisation and pruning, together with in-network computation, are discussed as ways to address communication cost. Computation-intensive inference may require task splitting across edge devices and servers, making communication efficiency especially important [2, 3].

2.6 Heterogeneity, Non-IID Data, and Stragglers

FL over edge nodes must address device heterogeneity, non-independent and identically distributed (non-IID) data, and limited communication resources. Synchronous FL can wait for the slowest worker, while asynchronous FL may cause frequent model transfers. The straggler effect can limit synchronous training time. Non-IID and imbalanced data can cause local updates to drift, resulting in instability and slower convergence; system heterogeneity can also lead to stragglers and dropouts. Participant selection is one proposed mitigation, while edge-assisted architectures, data augmentation, and mediator-based rescheduling are reported as approaches to non-IID bias. These challenges are part of the broader edge-intelligence research landscape [2].

2.7 Privacy and Security

FL does not eliminate privacy risks: a malicious participant may infer private data or poison a model. The surveyed literature discusses inference attacks, honest-but-curious servers, free-riding participants, and poisoning; differential privacy and other privacy-preserving approaches involve design trade-offs [2]. Specific attack-success rates are not reported here because the primary studies behind those figures have not been verified for this review.

2.8 Energy and Hardware Constraints

Latency, energy use, memory footprint, and available computing resources are relevant evaluation measures for edge workloads [2, 3]. Their measured values depend on the device, model, network conditions, and measurement setup. The cited DistrEdge evaluation, for example, does not consider memory constraints and reports its speedups only against the baselines and test cases used in that study [3]. A later implementation should therefore record these conditions and avoid direct comparison of results obtained under different protocols.

2.9 Application Domains

Reviewed applications include autonomous and connected vehicles, smart homes, industrial automation, healthcare, surveillance, real-time gaming, smart factories, and precision-agriculture pest detection.

Chapter 3

Methodology

Because the project remains at the learning and literature-review stage and the application is undecided, the methodology below is proposed rather than executed.

3.1 Review Protocol

This report presents a preliminary narrative review, not a systematic review. The team consulted online publication records and paper repositories for four core works discussed here: an edge-intelligence survey, a federated-learning study, a distributed edge-AI scheduling study, and a distributed CNN-inference study [2, 4, 1, 3]. A complete record of the databases and repositories searched was not maintained.

Exact search queries were not recorded. Terms recalled from the project's scope include "Distributed Edge AI," "Edge Intelligence," "Federated Learning," "Collaborative Inference," and "End-Edge-Cloud Computing," along with the titles "Distributed Edge AI Systems" and "DistrEdge: Speeding up Convolutional Neural Network Inference on Distributed Edge Devices." The work by McMahan et al. was also consulted through its publication record. These are reconstructed topic and title terms, not verified queries. Search dates were not recorded, and no formal publication-year restriction was documented. Publication years of works cited here describe the selected materials, not a predefined search range.

Formal inclusion and exclusion criteria, screening stages, and counts of papers initially considered were not recorded. The sources in this report were selected for relevance to distributed edge-AI architectures, edge intelligence, federated learning, collaborative inference, communication efficiency, resource heterogeneity, privacy and security, and energy and hardware constraints. These themes describe the review's scope rather than formal selection rules. The works cited are core references identified in the current draft, not necessarily the full set of sources consulted. Accordingly, the review does not claim exhaustive coverage or a reproducible systematic search.

A working extraction matrix has been prepared for the cited sources, recording their research focus, method, workload or dataset, hardware and network assumptions, evaluation metrics, reported findings, limitations, and relevance to the project. Details not stated by a source are marked as unreported or requiring further extraction. Search dates, actual queries, and screening decisions will be logged prospectively; missing historical search records will not be reconstructed.

3.2 Analytical Framework

Sources are synthesised across eight themes: definition and motivation; architecture; training; inference; communication; heterogeneity; privacy and security; and energy and hardware. This structure is intended to help map a later application choice to the relevant literature.

3.2.1 Six-Phase Research and Development Plan

The project is organized into six sequential but iterative phases.

Phase 1: Learning About Edge AI. Develop an understanding of Edge AI, Distributed Edge AI, end-edge-cloud architectures, distributed inference, federated learning, communication constraints, and resource management.

Phase 2: Literature Review and Reference Collection. Study relevant research papers, compare existing methods, record verified bibliographic details, and identify potential research gaps and application opportunities.

Phase 3: Small-Scale Experiments. First inventory available devices, software, and network connections, then select a candidate application and a local baseline. Compare it with one distributed configuration under recorded workload and network conditions. Measure end-to-end latency, throughput, communication volume, peak memory, and task accuracy; measure energy when suitable power-monitoring equipment is available.

Phase 4: Feasibility Assessment. Compare candidate applications according to technical relevance, implementation complexity, available resources, cost, and measurable evaluation criteria.

Phase 5: Project Implementation. Design and build a prototype for the application selected by the team following the preceding research and feasibility assessment.

Phase 6: Testing and Refinement. Evaluate the prototype against appropriate functional and performance requirements, analyze limitations, and refine the implementation.

The project is currently in Phases 1 and 2. The specific application, implementation platform, and final architecture have not yet been selected. Phases 3–6 therefore represent planned future work rather than completed activities. Figure 1.3 summarises all six phases and indicates the current stage.

3.3 Proposed Feasibility-Assessment Framework

A multi-criteria assessment is proposed using latency, energy efficiency, and computation and memory footprint, extended with bandwidth sensitivity, scalability, and cost. Candidate platforms could be compared using a reproducible benchmarking approach that

records the model, workload, device, software environment, and network conditions alongside latency, throughput, communication volume, memory, and power or energy where measurable. Any acceptance thresholds will be set only after the application requirements and available resources are confirmed. The candidate-selection process and criteria are summarised in Figure 3.1.

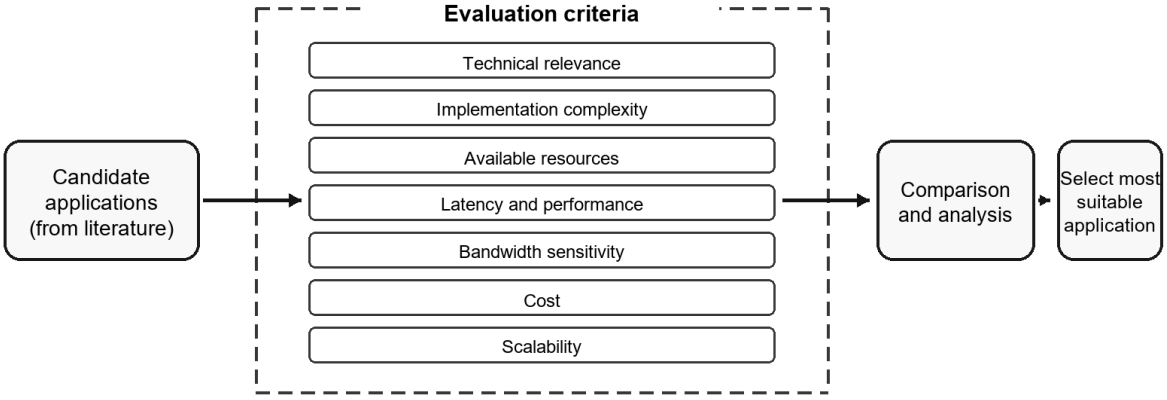


Figure 3.1: Proposed Feasibility-Assessment Framework.

3.4 Proposed Technical Methodology

Two possible implementation directions are proposed for a later phase:

1. **Partitioned collaborative inference:** split a model by layer or horizontally across devices and exchange intermediate outputs, following published partitioning schemes [3].

2. **Federated training:** participants train local models and share model updates through an aggregation server, following federated averaging and its evaluation on distributed, unbalanced data [4].

Neither direction has been implemented; both remain proposals for subsequent work.

The approaches described above are preliminary candidates identified from the literature, not finalized implementation choices. Their suitability will be investigated through further literature analysis, small-scale experimentation, and feasibility assessment. The final approach will be selected based on the identified problem, available resources, implementation requirements, and evaluation objectives.

Before choosing between these directions, the team will document its available hardware, software, datasets, and network setup; select a workload and local baseline; and check that the chosen evaluation measures can be collected on that setup. The comparison will treat federated training and partitioned inference as different problem types, not interchangeable implementations.

Chapter 4

Results and Discussion

As this is a preliminary, literature-stage study, the results are the outcomes of the review and feasibility synthesis rather than experimental findings.

4.1 Literature Synthesis

The reviewed literature converges on layered end-edge-cloud architectures, with federated training and partitioned or collaborative inference as two major technical families [2, 3]. Communication efficiency is a recurring bottleneck across both. Table 4.1 compares the four selected works by focus, method, evaluation, and reported limitations. Numerical results are presented as source-specific findings, not as directly comparable measurements. The comparison also indicates how each work can inform a later project decision.

Table 4.1: Comparison of selected literature, its evidence, and project relevance.

Study	Contribution and method	Evaluation and finding	Limitation and project relevance
Hu et al. 2023 [1]	Dynamic scheduling for a containerized video-detection pipeline across heterogeneous edge nodes.	Raspberry Pi, MacBook Pro, and Asus nodes; varied video quality and container count. Parallelism helped to a threshold; further containers increased processing time.	One application and deployment. Informs task-placement and communication-cost evaluation.
Deng et al. 2020 [2]	Survey taxonomy distinguishing AI for edge from AI on edge.	Literature synthesis; no experimental benchmark.	Scope-dependent; provides conceptual context, not a performance baseline.
Hou et al. 2022 [3]	DistrEdge uses deep reinforcement learning to partition CNN inference across heterogeneous devices.	Jetson devices under varied device and network conditions; reports 1.1–3× speedup against tested baselines.	Memory is not modeled. Candidate inference approach; measure memory in future tests.
McMahan et al. 2017 [4]	Federated averaging aggregates locally computed model updates.	Five architectures and four datasets; reports 10–100× fewer rounds than synchronized SGD in tested settings.	Training-focused; workload, data partition, and baseline matter. Not an inference comparison.

The works inform different decisions rather than providing directly comparable alternatives: Deng et al. provide a survey framework; McMahan et al. provide a federated-training baseline; Hou et al. evaluate partitioned CNN inference; and Hu et al. study task placement in a video-processing application [2, 4, 3, 1]. The reported metrics and test conditions differ, so their quantitative results should not be ranked against one another.

4.2 Feasibility Indications

The selected studies indicate that performance depends on how tasks are partitioned, the capabilities of participating devices, and the communication conditions. Hu et al.

report diminishing returns when additional detection containers are added to their video application, whereas Hou et al. report a 1.1–3× speedup only against the baselines in their own CNN-inference evaluation [1, 3]. These findings illustrate different workloads and evaluation setups and should not be interpreted as a direct comparison. The included Deng et al. paper is a survey and provides context rather than new benchmark results [2].

4.3 Discussion and Limitations

The synthesis indicates a trade-off between latency and energy on one side and communication cost on the other. Moving computation closer to the data may reduce latency and bandwidth pressure, but lower edge computing capacity and device heterogeneity can require model partitioning, compression, or collaborative execution. Non-IID data and slow or unreliable nodes can create stragglers and dropouts. Privacy and security risks, including inference and poisoning attacks, also constrain how data and model updates are handled.

No implementation or measurements have been carried out; this feasibility assessment relies entirely on published results. Those results use different platforms, workloads, and measurement setups. Even power-measurement protocols vary, so direct cross-study comparison of performance and energy values must be treated cautiously. The field is still developing, and the definition of "edge" remains unsettled. Open challenges include resource constraints, security threats, scalability, and communication overhead.

Chapter 5

Conclusion and Future Work

This preliminary report has reviewed distributed edge AI and assessed feasibility at a conceptual level. The literature describes a shift from centralised cloud AI toward layered end-edge-cloud architectures, with federated learning and collaborative DNN partitioning as major technical families. Potential suitability for latency- and privacy-sensitive applications is bounded by communication cost, device and data heterogeneity, energy constraints, and privacy and security risks [2, 3, 4].

The project is currently in the foundational learning and literature-review stages. Immediate milestones are to complete a source-extraction matrix, inventory the available hardware, software, datasets, and network connections, and select a candidate workload and local baseline. The team can then run a small pilot, recording latency, throughput, communication volume, memory, accuracy, and energy where measurable before deciding whether federated training or partitioned inference is feasible for that workload. Acceptance thresholds will be set after the application and its requirements are confirmed. No implementation or experimental results are claimed at the present stage.

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